

Entity-Relationship model

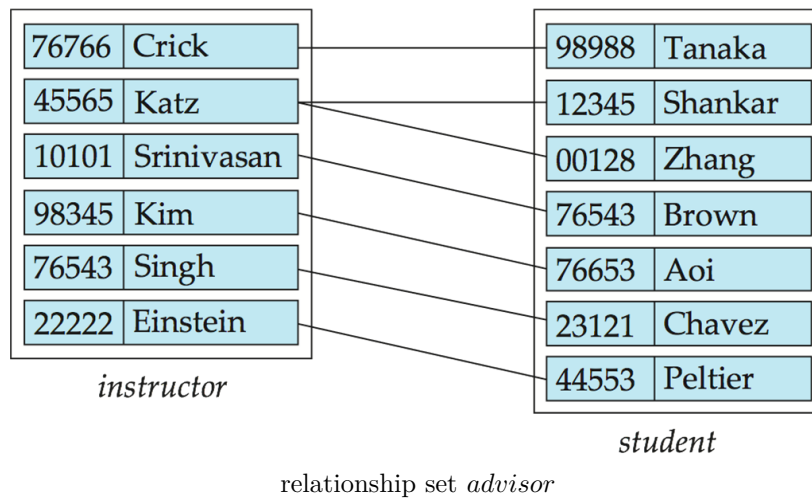
Weihai Yu

October 17, 2019

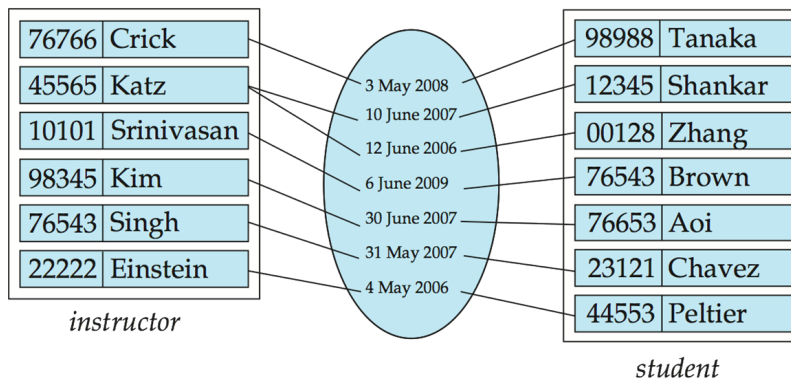
Entity and entity set

- an **entity** is a “thing” with attributes
- attributes can be composite or multivalued
- an **entity set** is a set of entities of the same type
- **keys**, similar to the relational data model

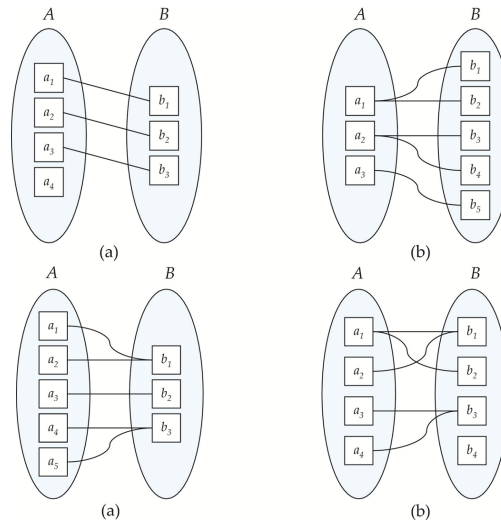
Relationship and relationship set



Relationship set with attribute

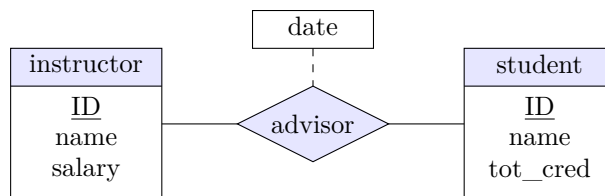


Mapping cardinalities, participation constraints

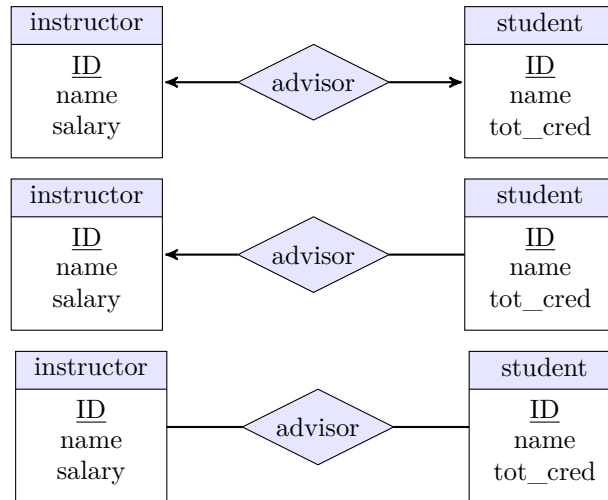


cardinalities: one to one, one to many, many to one, many to many
 participation: partial, total

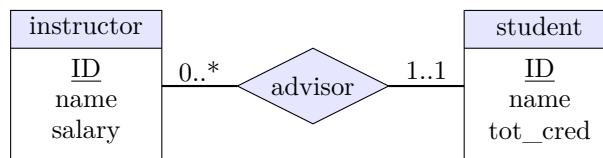
ER diagram



Cardinalities in ER diagram



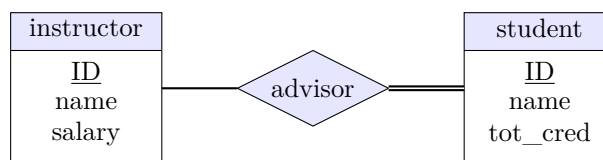
Cardinalities in ER diagram



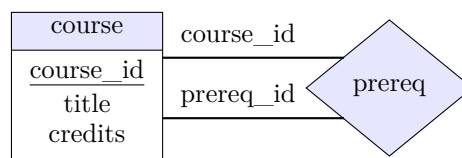
Is the *advisor* relationship set one to one, one to many, many to one or many to many?

one to many

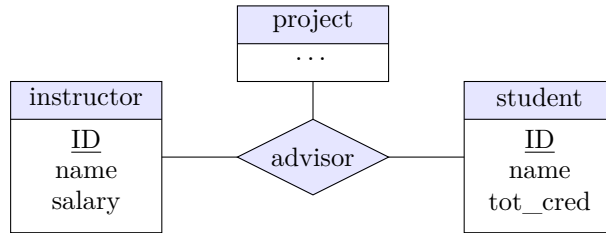
Total participation



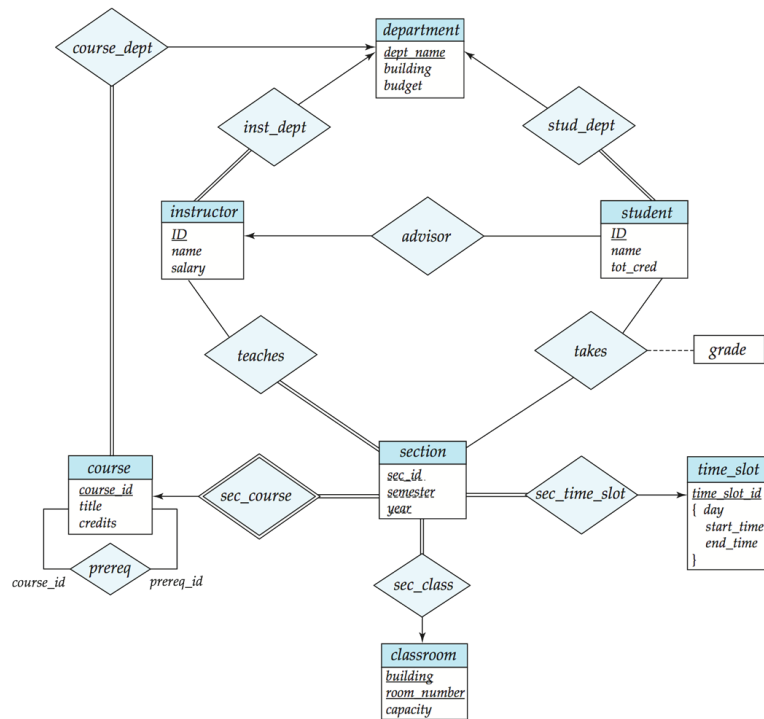
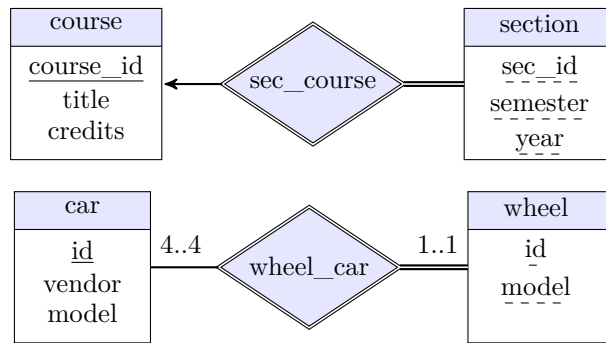
Roles ER diagram



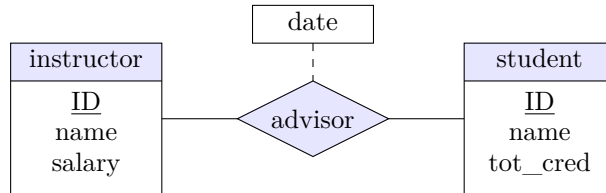
Ternary relationship



Weak entity set

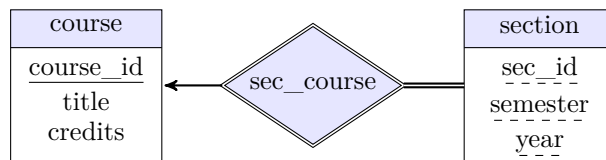


ER model to relation schemas



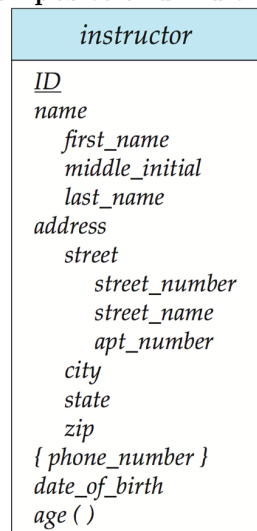
- *instructor*(ID, name, salary)
- *student*(ID, name, tot_cred)
- *advisor*(inst_id, stud_id, date)

Representing weak entity sets



section(course_id, sec_id, semester, year)

Composite and multivalued attributes



instructor(ID, first_name, ...,
street_name, ..., zip,
data_of_birth, age)
inst_phone(ID, number)